06_missing_index_dmvs

Propietario	© Carlos Roberto
Etiquetas	

6.0 GENERATE MISSING INDEX SIGNAL

```
USE STN_Lab;
GO

-- ==========================================================
==
-- 6.0 -- GENERATE MISSING INDEX SIGNAL
-- Run queries that trigger missing index recommendations.
-- DMVs only populate after queries execute without a
-- suitable index -- cold DMVs produce no results.
-- ==========================================================
==

SET STATISTICS IO ON;

SELECT ContribuyenteID, NIT, RazonSocial, DepartamentoID, Tip
oPersona
FROM dbo.Contribuyente
WHERE DepartamentoID = 1
  AND TipoPersona = 'J';

SELECT ContribuyenteID, NIT, RazonSocial
FROM dbo.Contribuyente
WHERE TipoPersona = 'N'
  AND Estado = 'A';

SELECT DeclaracionID, ContribuyenteID, PeriodoFiscal, MontoIm
```

```
puesto
FROM dbo.Declaracion
WHERE FechaPresentacion >= '2023-01-01'
  AND TipoImpuestoID = 1;

SELECT DeclaracionID, ContribuyenteID, MontoDeclarado, Estado
FROM dbo.Declaracion
WHERE Estado = 'P'
  AND PeriodoFiscal = '2023';

SET STATISTICS IO OFF;
GO
```

6.1 RAW MISSING INDEX RECOMMENDATIONS

```
SELECT
    OBJECT_NAME(mid.object_id)          AS table_name,
    mid.equality_columns,
    mid.inequality_columns,
    mid.included_columns,
    migs.user_seeks,
    migs.user_scans,
    migs.avg_total_user_cost,
    migs.avg_user_impact
FROM sys.dm_db_missing_index_details mid
JOIN sys.dm_db_missing_index_groups mig
    ON mid.index_handle = mig.index_handle
JOIN sys.dm_db_missing_index_group_stats migs
    ON mig.index_group_handle = migs.group_handle
WHERE mid.database_id = DB_ID()
ORDER BY migs.avg_total_user_cost * migs.avg_user_impact DES
C;
GO
```

OBSERVACION 6.1
Anotar cuantas sugerencias aparecen y para que tablas.

Results | Messages

	table_name	equality_columns	inequality_columns	included_columns	user_seeks	user_scans	avg_total_user_cost	avg
1	Declaracion	[TipoImpuestoID]	[FechaPresentacion]	[DeclaracionID], [ContribuyenteID], [PeriodoFisc...	1	0	11.0842248432899	89
2	Contribuyente	[TipoPersona], [DepartamentoID]	NULL	[NIT], [RazonSocial]	1	0	1.36668940740741	90

Query executed successfully. | c13\lab_db (16.0 RTM) | carlos_roberto (53) | STN_Lab | 00:00:00 | Row: 1, Col: 1 | 2 rows

6.2 PRIORITIZED MISSING INDEX REPORT

Impact score = user_seeks x avg_cost x avg_impact / 100. Solo implementar indexes con score mayor a 1,000.

```
SELECT
    OBJECT_NAME(mid.object_id)                          AS ta
ble_name,
    mid.equality_columns,
    mid.inequality_columns,
    mid.included_columns,
    migs.user_seeks,
    CAST(migs.avg_user_impact AS DECIMAL(5,1))          AS im
pact_pct,
    CAST(
        migs.user_seeks
        * migs.avg_total_user_cost
        * migs.avg_user_impact / 100.0
    AS DECIMAL(18,2))                                   AS pr
iority_score,
    'CREATE NONCLUSTERED INDEX [IX_Missing_' +
        CAST(mig.index_group_handle AS VARCHAR) + '] ON ' +
        mid.statement + ' (' +
```

```sql
        ISNULL(mid.equality_columns, '') +
        CASE
            WHEN mid.equality_columns IS NOT NULL
             AND mid.inequality_columns IS NOT NULL THEN ', '
            ELSE ''
        END +
        ISNULL(mid.inequality_columns, '') + ')' +
        ISNULL(' INCLUDE (' + mid.included_columns + ')', '')
    AS suggested_index
FROM sys.dm_db_missing_index_details mid
JOIN sys.dm_db_missing_index_groups mig
    ON mid.index_handle = mig.index_handle
JOIN sys.dm_db_missing_index_group_stats migs
    ON mig.index_group_handle = migs.group_handle
WHERE mid.database_id = DB_ID()
  AND migs.user_seeks > 0
ORDER BY priority_score DESC;
GO
```

Results | Messages

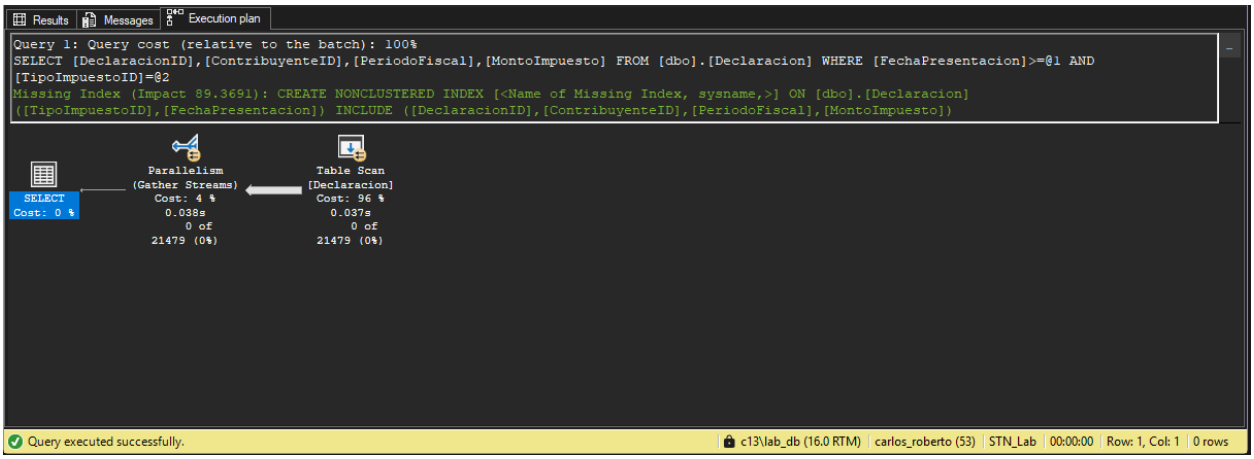
	table_name	equality_columns	inequality_columns	included_columns	user_seeks	impact_pct	priority_score	suggested_index
1	Declaracion	[TipoImpuestoID]	[FechaPresentaci...	[DeclaracionID], [ContribuyenteID], [PeriodoFis...	1	89.4	9.91	CREATE NONCLUSTERED INDEX [IX_Missing_4] ON [ST...
2	Contribuyente	[TipoPersona], [DepartamentoID]	NULL	[NIT], [RazonSocial]	1	90.3	1.23	CREATE NONCLUSTERED INDEX [IX_Missing_2] ON [ST...

Query executed successfully. | c13\lab_db (16.0 RTM) | carlos_roberto (53) | STN_Lab | 00:00:00 | Row: 1, Col: 1 | 2 rows

6.3 IMPLEMENTAR LA RECOMENDACIÓN CON MAYOR SCORE

6.3.1 Antes de implementar

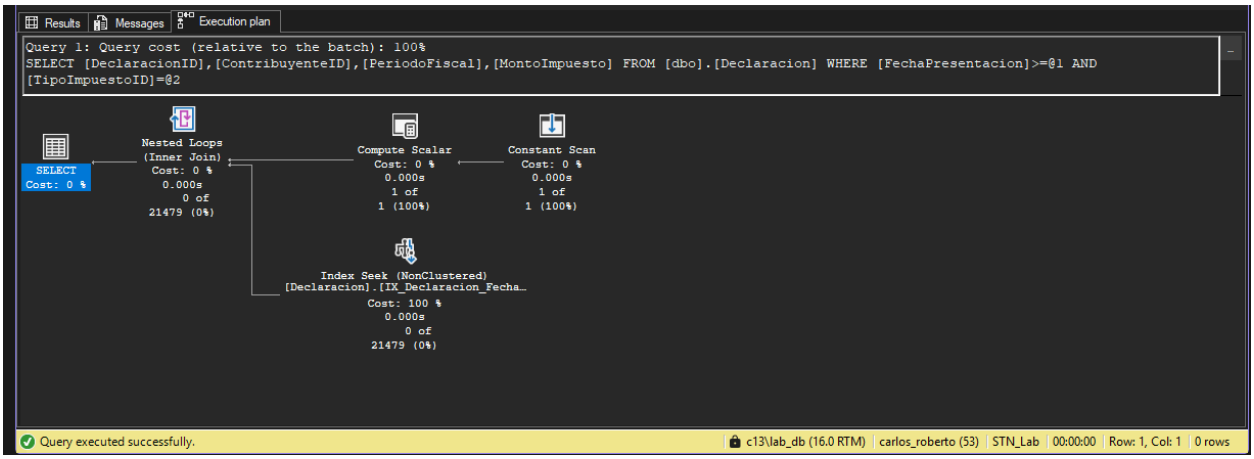
```sql
SET STATISTICS IO ON;

SELECT DeclaracionID, ContribuyenteID, PeriodoFiscal, MontoIm
puesto
FROM dbo.Declaracion
WHERE FechaPresentacion >= '2023-01-01'
  AND TipoImpuestoID = 1;

SET STATISTICS IO OFF;
GO
```



6.3.2 Crear el indice sugerido (ajustar segun resultado de 6.2)

```sql
CREATE NONCLUSTERED INDEX [IX_Declaracion_Fecha_TipoImpuesto]
ON [STN_Lab].[dbo].[Declaracion] ([TipoImpuestoID],
[FechaPresentacion]) INCLUDE ([DeclaracionID],
[ContribuyenteID], [PeriodoFiscal], [MontoImpuesto])
```

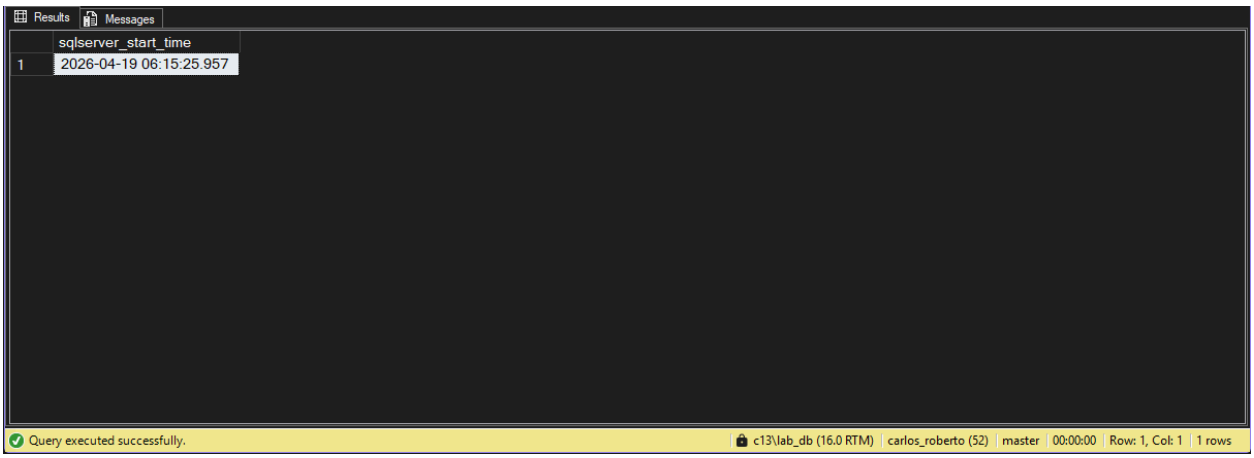
6.3.3 Re-ejecutar el mismo query de 6.3.1



6.4 LIMITACIONES DE LOS DMVs

Los DMVs se resetean con cada reinicio del servicio. No reflejan el historico real de produccion. Una sugerencia con score alto en lab puede ser irrelevante en produccion si el query no se ejecuta frecuentemente.

```
SELECT
    sqlserver_start_time
FROM sys.dm_os_sys_info;
GO
```



OBSERVACION 6.4

- Anotar desde cuando están corriendo los DMVs.
- Si el servicio se reinicio recientemente los datos son parciales.

6.5 CLEANUP

```
USE STN_Lab;
GO
DROP INDEX IF EXISTS IX_Declaracion_Fecha_TipoImpuesto ON db
o.Declaracion;
```